

PREFACE

On behalf of the Organizing and Programme Committee I, Prof. Srikanta Patnaik, General Chair of the conference, welcome all the participants to the 10th Annual International Conference of Information and Communication Technology (ICICT2020) scheduled during April 24–25, 2020 but held during November 13–15, 2020 at Wuhan, China. This annual conference is being organized each year by Interscience Research Network, an international professional body, in association with International Journal of Information and Communication Technology, published by Inderscience Publishing House. Like every edition, this edition of ICICT–2020 was academically very rich and we had two eminent professors as Key Note Speakers namely Prof. Hamido FUJITA, Director of Intelligent Software Systems Iwate Prefectural University, Japan and Prof. Dr. Roumen Kountchev, Faculty of Telecommunications, Department of Radio Communications and Video Technologies at the Technical University of Sofia, Bulgaria.

There is no doubt that Information and Communication Technology has established its profound influence on the day-to-day operation, functioning and decision making of modern organizations. New technologies are constantly emerging, which are enabling applications in various domains and services. Information Technology is no longer a functional area within an organization, but is integral to the functioning of an organization. In the recent time, Information and Communication Technology (ICT) is probably the single most important facilitator of the integration process. The strong influence of Information and Communication Technology in shaping the process of globalization, particularly in the productive, commercial and financial spheres, is also widely recognized. Creating economic opportunities and contributing to poverty reduction is another thrust area for the emerging epoch of ICT.

This edition of ICICT covered the following areas but not limited to: Sensor/Ad hoc Networks, Information Security, Human–Computer Intelligent Interaction, IT and Network Security, Computational linguistics, Internet and Web Based Systems and Products, Distributed and Cooperative Media, Data Mining and Data Warehousing, Interactive Communication Media and Contents, Computer Network Planning And Design, Social Interaction Computer Network Administration, Mobile Communication, Signal Representation and Processing, Digital Speech and Image Processing, Biomedical Applications E-governance, Rural Applications of IT, Networking–Enterprise Applications ERP/CRM/ SCM, Computer Control of Industrial Processes, Microprocessor Interfacing and Hardware Design, ICT Act and Cyber Law, Re-engineering ICT

infrastructure Standards and Security, Virtual Reality Artificial Intelligence, ICT Application in Agriculture and Aids for the Disabled. This volume covers various articles covering the recent developments of Information and Communication Technology categorized into five (5) tracks, such as follows:

- 1 IOT and Cloud Computing
- 2 Machine Learning and Machine Intelligence
- 3 Commutation and Networking
- 4 Image Processing and Computer Vision
- 5 Big Data and Analytics and
- 6 ICT Applications

ACKNOWLEDGEMENT

Due to the pandemic, this edition of ICICT-2020, was arranged virtually, however, this was attended by more than 200 participants across globe and 119 papers were shortlisted and published in this proceedings. The papers covered in this proceeding are the result of the efforts of the researchers working in various domains of Information and Communication Technology. We are thankful to the authors and paper contributors of this volume.

We are thankful to the Editor-in-Chief and the anonymous review committee members of the ELSEVIER Journal of Procedia Computer Science, for their support to bring out the proceedings of 10th International Conference of Information and Communication Technology. It is noteworthy to mention here that this was really a big boost for us to continue this conference series.

We are thankful to our friends namely Prof. Hamido FUJITA, Director of Intelligent Software Systems Iwate Prefectural University, Japan and Prof. Dr.Roumen Kountchev, Faculty of Telecommunications, Department of Radio Communications and Video Technologies at the Technical University of Sofia, Bulgaria, for their key note address. We are also thankful to the experts and reviewers who have worked for this volume despite of the veil of their anonymity.

We shall not forget to inform you that the next edition of the conference i.e. 8th International Congress of Information and Communication Technology (ICICT2021) will take place at Wuhan, China during November 13–15, 2021 and we shall be updating you regarding the dates and venue of the conference.

It was really a different experience to interact with the scholars and researchers virtually.

I am sure that the readers shall get immense ideas and knowledge from this volume of Procedia Computer Science.

Prof. Srikanta Patnaik
General Chair: ICICT-2020